IFN647 Week 11 Review Questions
Solutions

Q1 (**Distance Function**)

Table 1 shows an example of a binary information table.

Document	t_1	t_2	t_3	t_4	t_5	t_6	t_7
d_1	1	1	0	0	0	0	0
d_2	0	0	1	1	0	1	0
d_3	0	0	1	1	1	1	0
d_4	0	0	1	1	1	1	0
d_5	1	1	0	0	0	1	1
d_6	1	1	0	0	0	1	1

Table 1. A binary information table

(1) Draw the binary contingency table of Table 1 for comparing documents d_1 and d_2.
(2) Evaluate the Jaccard distance between d_1 and d_2.
(3) Evaluate the Simple matching coefficient distance between d_1 and d_2.

Solution:
(1)

		Document d_2		
		1	0	Sum
Document d_1	1	0	2	2
	0	3	2	5
	Sum	3	4	7

(2)
disJac(*d*1, *d*2) = (r+s) / (q+r+s) = 5/5 = 1

(3)
disS(*d*1, *d*2) = (r+s) / (q+r+s+t) = 5/7

Q2 (Cluster Precision)

Table 2 shows a collection of documents, which includes 10 documents (each with a unique document id) $D = \{d_1, d_2, \ldots, d_{10}\}$, two classes (spam and not spam) *Class* = {'spam', 'not_spam'}, and a vocabulary $V = \{w_1, w_2, \ldots, w_5\}$ = {'cheep', 'buy', 'banking', 'dinner', 'the'}.

document *id*	cheap	buy	banking	dinner	the	*class*
1	0	0	0	0	2	not spam
2	3	0	1	0	1	spam
3	0	0	0	0	1	not spam
4	2	0	3	0	2	spam
5	5	2	0	0	1	spam
6	0	0	1	0	1	not spam
7	0	1	1	0	1	not spam
8	0	0	0	0	1	not spam
9	0	0	0	0	1	not spam
10	1	1	0	1	2	not spam

Table 2. A collection of documents

Assume your cluster algorithm produces two (K=2) clusters $C_1 = \{d_2, d_5, d_{10}\}$ and $C_2 = \{d_1, d_3, d_4, d_6, d_7, d_8, d_9\}$.

(1) Calculate $MaxClass(C_i)$ for i = 1 and 2.
(2) Calculate the cluster precision of your cluster algorithm.

Solution:

(1)
Let $C\_spam = \{d_2, d_4, d_5\}$, i.e., for any d in C_spam, $label(d)$ = 'spam', and $C\_not\_spam = \{d_1, d_3, d_6, d_7, d_8, d_9, d_{10}\}$, i.e., for any d in C_not_spam, $label(d)$ = 'not_spam'.

$label(C_1)$ = 'spam', $label(C_2)$ = 'not_spam'.
$MaxClass(C_1) = \{d_2, d_5\}$, $MaxClass(C_2) = \{d_1, d_3, d_6, d_7, d_8, d_9\}$.

(2)
$Cluster_{precision} = (2 + 6) / 10 = 0.8$

Q3 (Recommender systems)

Solution:

```
# The common similarity measures used for clustering users is the correlation measure.
# Typically, users are represented by their rating vectors of items.
# We use a list of lists Rvs to represent rating vectors for all users
# For example
#   I = [0,1,2,3]  # 4 items in I numbered from 0 to 3 – a list
#   U = [0,1,2,3,4] # 5 users in U numbered from 0 to 4 – a list
#   Rvs = [[1,2,1,0], [0,1,1,1], [2,1,3,5], [1,0,2,0], [0,2,3,4]] – a list of lists
#      5 users' rating vectors for 4 items, where 0 means the user has not yet rated the item.

def my_correlation(Rvs):
        U = [i for i in range(len(Rvs))]
        I = [i for i in range(len(Rvs[0]))]
        Uc = [[1 for i in range(len(U))] for i in range(len(U))]
        …
        return(U, I, Uc)
# It returns U, I and Uc (the correlation matrix, a list of lists)

def my_cluster(Rvs):
    …
    return C # clusters – list of user lists

# The predicted rating to unseen items for all users
def my_prediction(Rvs, C, U, I):
        …
        return Rvs   # updated Rvs
```